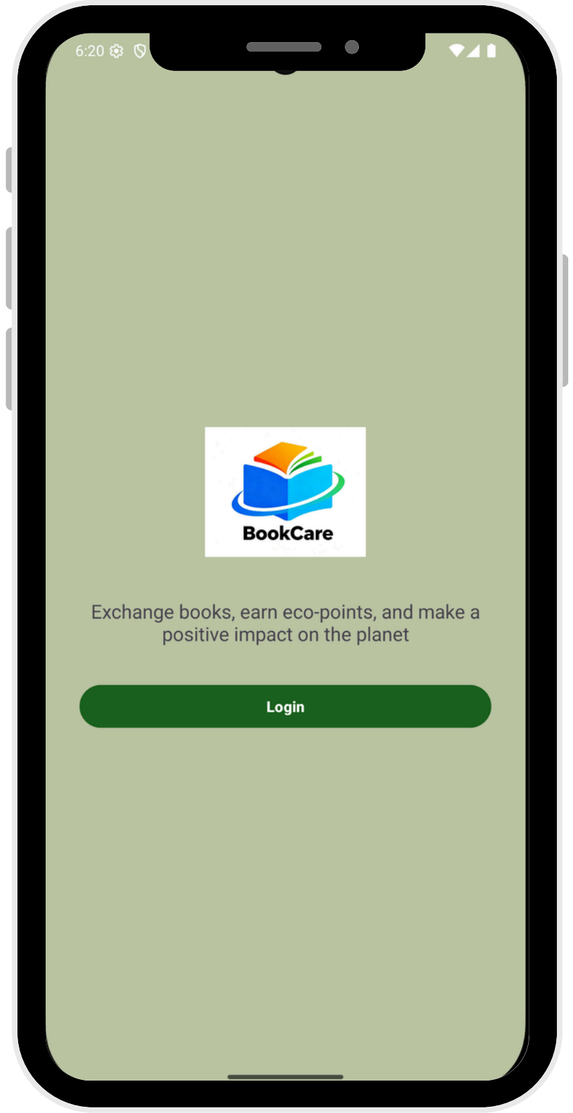


BookCare - Sustainable Book Reuse Mobile Application

WIA2007/WIX3004
Mobile Application Development
Semester 1, Session 2025/2026

Occurrence: 6

Lecturer: Dr. Mohamad Hazim Bin Md Hanif

Team Member: Lee Zi Xuan, Chong Chee Yan, Ho Kang Zheng, Lee Yu Xuan, Lee Qian Yi, Yeong Hui Ni

SDG 12 – Responsible Consumption & Production

SDG 12: **Responsible Consumption & Production**

Target 12.5:
Reduce waste generation through prevention, reduction, recycling and reuse.

Problem Addressed: **Book Waste**

- Physical books are often discarded after limited use
- Lack of structured platforms for book reuse and exchange

Our Solution: **BookCare**

- Enables **donation & exchange** of second-hand books
- Encourages reuse via **eco-points & badges**
- Builds a **sustainable reading community**

Our demonstration will highlight key modules that showcase BookCare's functionality and its commitment to sustainable practices. Each feature directly supports our functional requirements.

01

User Authentication & Profile

Secure access and personalized user settings.

02

Book Marketplace

Browse, upload, and manage book listings.

03

Book Recommendation System

Personalized book suggestions for users.

04

Smart Donation & Free Exchange

Streamlined process for giving and receiving books.

05

Community & Review Forum

Interactive platform for discussions and reviews.

06

Eco-Impact & Badge System

Gamified incentives for sustainable actions.

DEMO: User Authentication & Profile

Secure Access

Seamless registration and login using email and password, with a robust “Forgot Password” recovery option.

Personalized Experience

Users can easily edit their profile and set reading preferences to tailor their app experience.

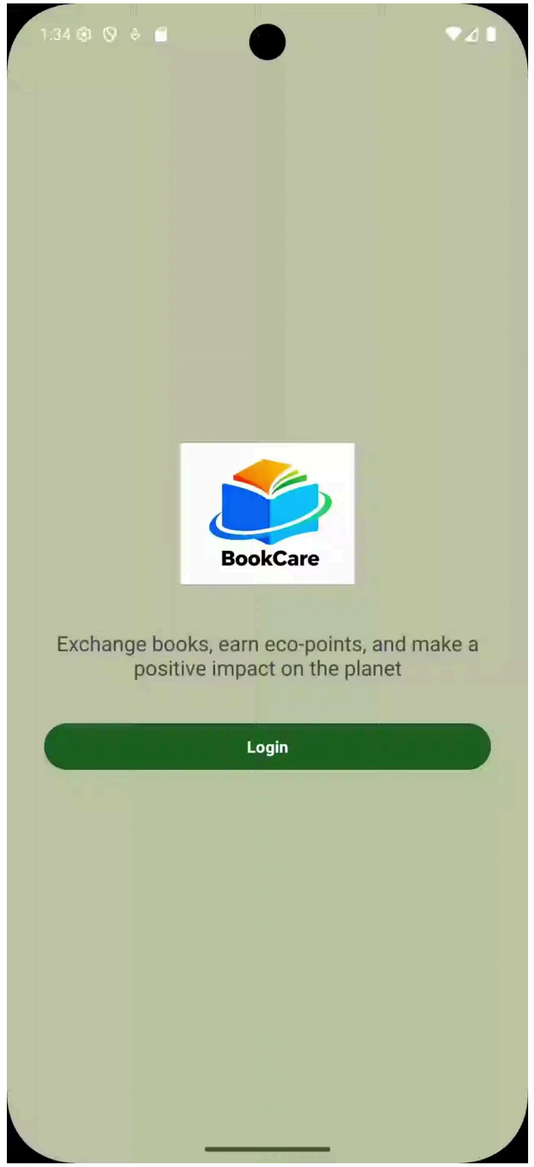
Customizable Interface

A unique Green/Blue toggle allows users to switch between eco-friendly and classic themes.

Privacy Control

Secure logout functionally ensures user data protection.

Contribution to SDG 12.5: Building a Trusted Community



DEMO: Book Marketplace

Upload Book Listing

Users upload book details and choose Exchange or Donate listing types.

Browse Books

Users browse available books categorized by listing type

Search Books

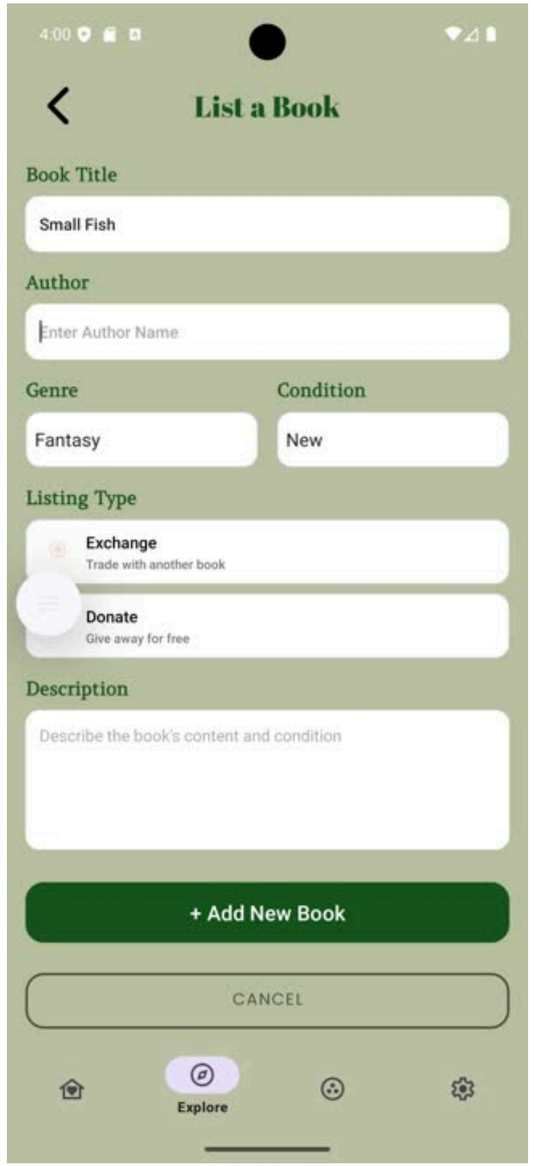
Users search the books by title or author for faster discovery.

Listing Management

Users can manage their own listings, update book details, and monitor exchange or donation requests.

View Book Details

Users can view the book condition, owner details, and listing status.



Contribution to SDG 12.5: Minimizes Waste via Book Exchange and Donation

DEMO: Smart Donation & Free Exchange

Description:

Smart Exchange & Donation Flow

Clear request, confirmation, and status tracking ensure smooth and transparent transactions between users.

Donate Book

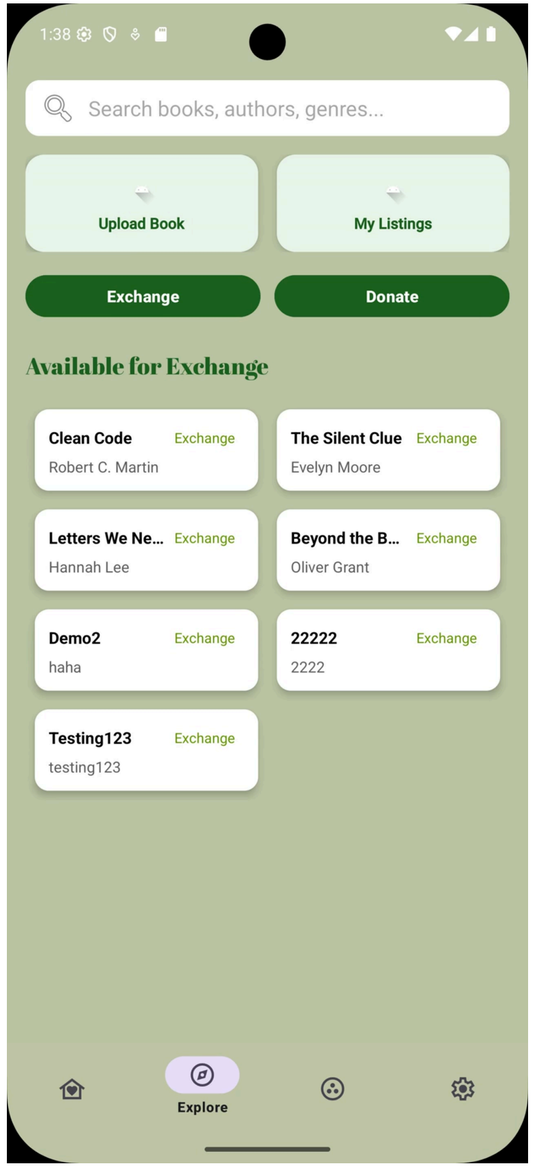
Users list books as free donations for others to claim.

Request Exchange

Users send or accept book exchange requests.

Confirm Exchange

The system confirms exchanges and updates listing status.



Contribution to SDG 12.5: Minimizes Waste via Book Exchange and Donation

DEMO: Eco-Impact & Badges System

Automated Eco-Points Calculation

Eco-points are automatically awarded when users donate or exchange books, reinforcing sustainable behavior.

Impact Tracking

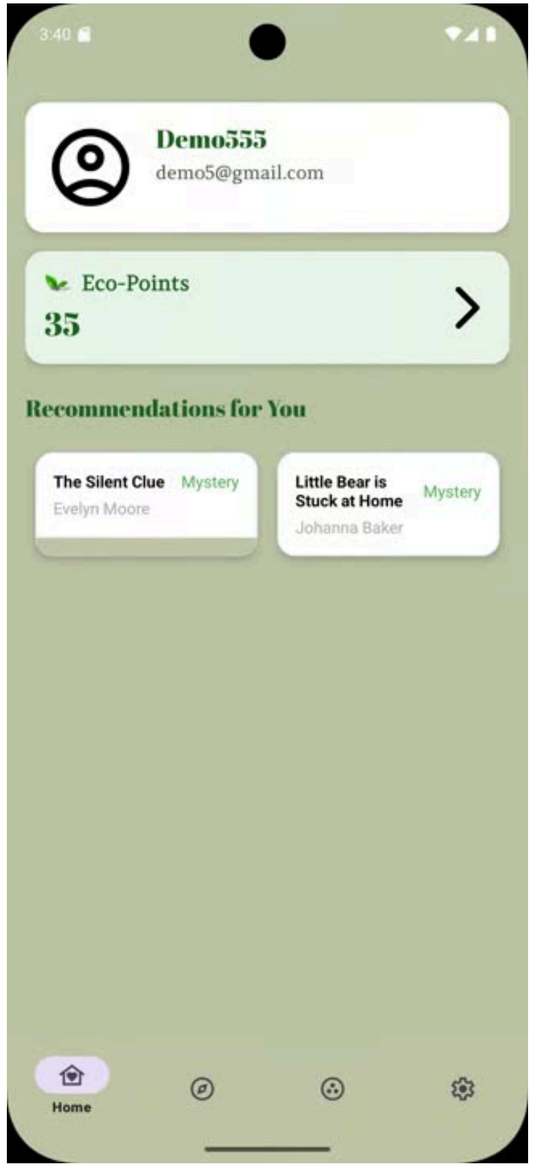
Users can view their eco-points history to understand their contribution to environmental sustainability.

Positive Reinforcement

Immediate visual feedback confirms successful actions and rewards users instantly.

10
Points for Donation

5
Points for Exchange



Contribution to SDG 12.5: Encouraging Responsible Consumption through Reward-based Sustainability

DEMO: Book Recommendation System

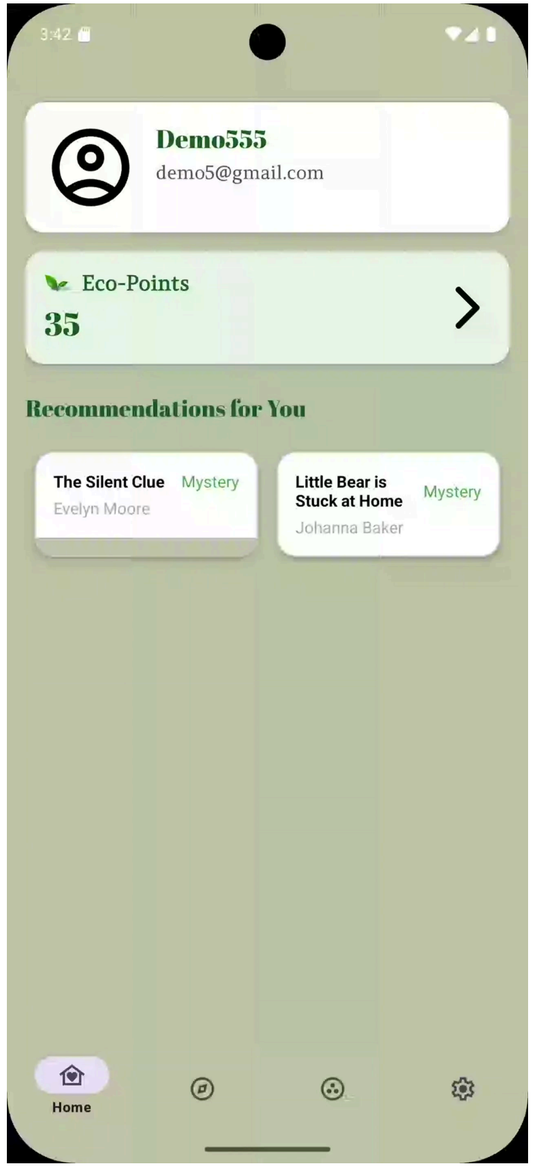
Description:

Personalized Recommendations

- **Smart Filtering:** Filters by genre chosen by users.
- **Circular Economy:** Prioritizes pre-loved books to reduce waste.
- **Adaptive Learning:** The more you use it, the better the picks get.

How It Works:

→ → →



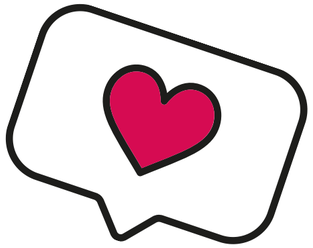
Contribution to SDG 12.5: Supporting Responsible Consumption by Promoting Relevant Second-hand Books based on User Preferences

DEMO: Community & Review Forum

Description:

Interactive Community Forum

Users can post reviews, engage in discussions, and like/comment on posts, fostering a dynamic environment.

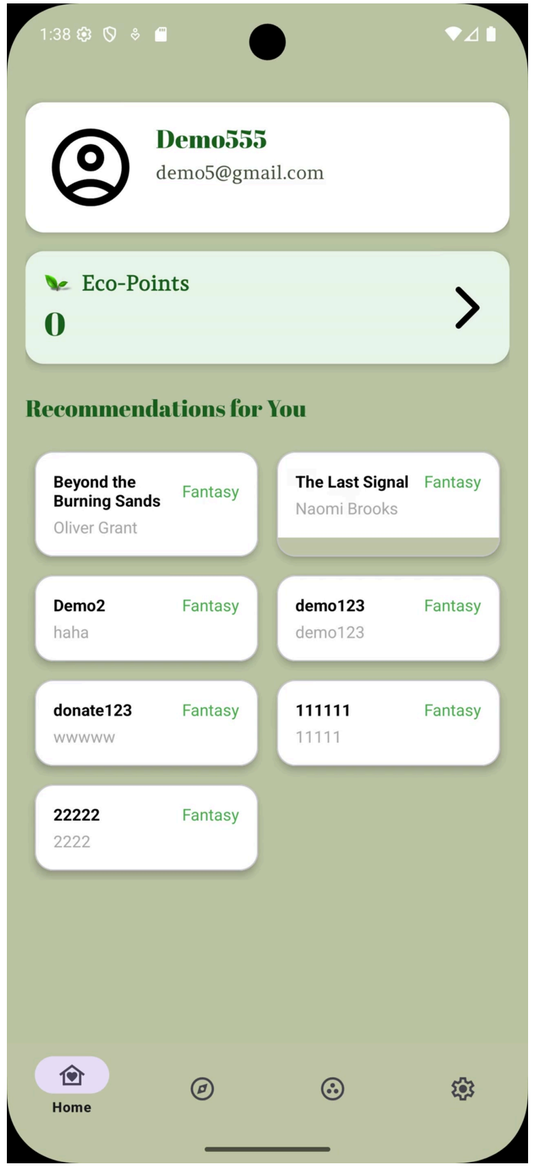


Community & Review Forum

Users can share reviews, post discussions, and interact with others to build trust.

Social Engagement

Likes and comments foster a sense of community and shared responsibility.



Contribution to SDG 12.5: Build Social Trust and Transparency within the App.

Design & Usability Considerations

UI / UX Principles

- **Simplicity:** Streamlined workflows with minimal steps for intuitive user experience.
- **Consistency:** Uniform design elements including buttons, icons, and typography.
- **Accessibility:** Offering Green/Blue modes to toggle.
- **Feedback:** Providing clear visual confirmations and prompts for user actions.

Color Schemes & Layout

- Green theme (#BAC4A3)
- Blue mode (#B0DEF5)
- Dark Green key actions & button (#7C8A67)
- Card-based layouts & Grid views

Interaction Flows

- Navigation between major features
- Core actions (upload, exchange, donation, discussions)
- Visual feedback (confirmation messages, eco-points updates)

Technical Implementation [Development Tools & Framework]

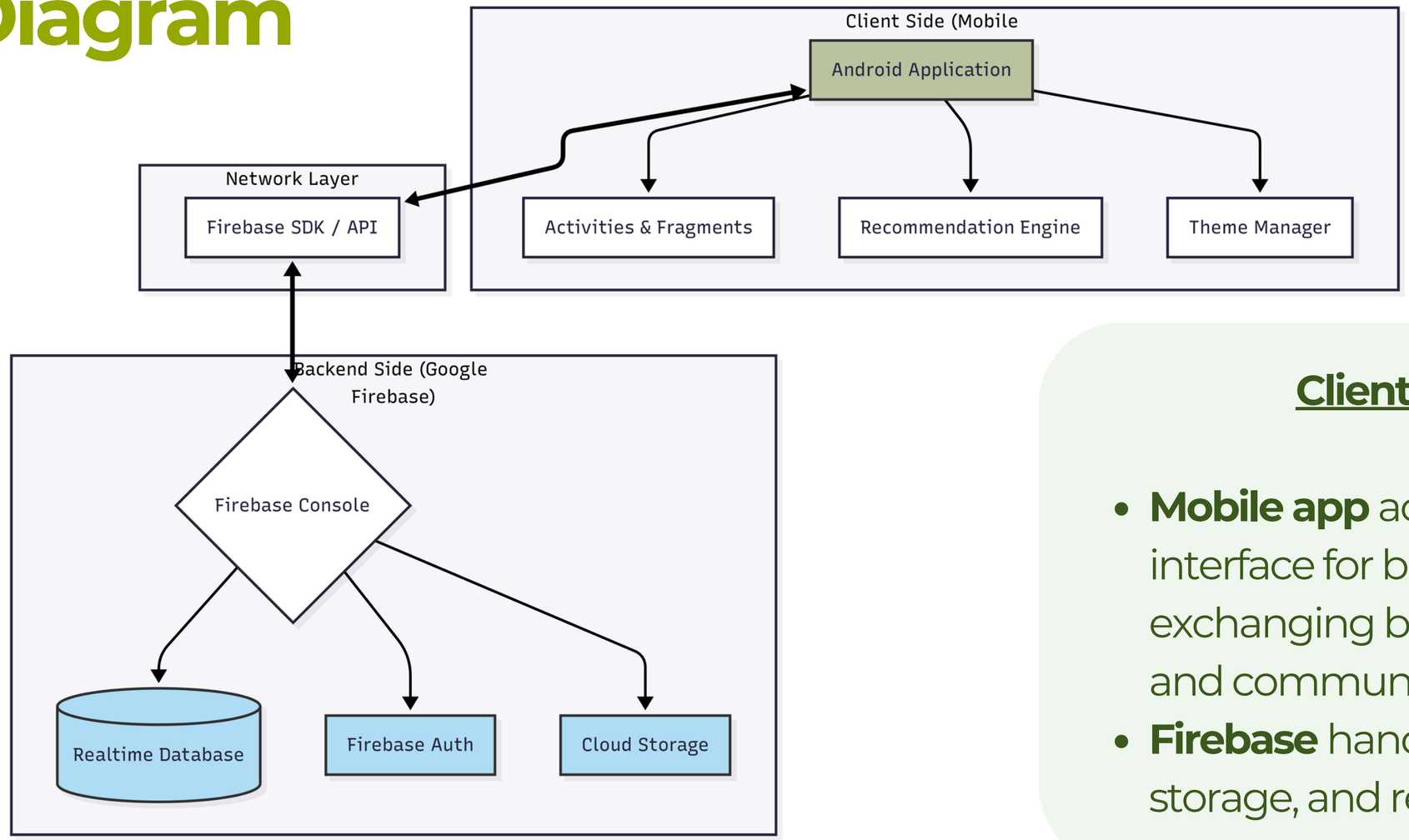
Android Studio
Primary IDE for robust mobile application development.

Firebase Authentication
Secure and scalable user authentication services.

Firebase Realtime Database
Real-time data synchronization with NoSQL, JSON-based storage for dynamic content.

Java
Implementing application logic and UI interactions.

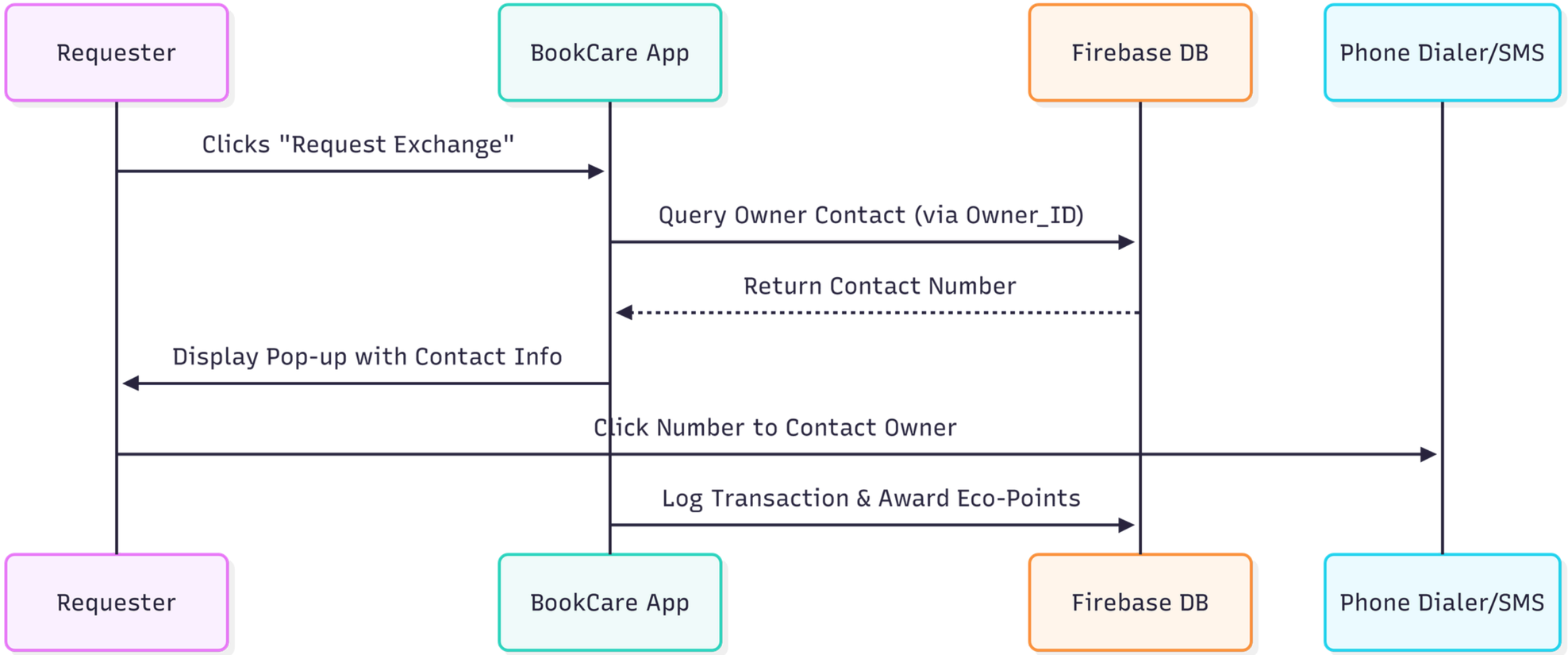
System Architecture Diagram



Client-Server Model

- **Mobile app** acts as the **client** (user interface for browsing, uploading, exchanging books, recommendations, and community interaction)
- **Firebase** handles authentication, data storage, and real-time updates

User Exchange Process



Data Management Approach

- **NoSQL JSON-based structure** for users, books, transactions, and posts
- **Real-time synchronization** ensures updates are reflected instantly across users
- Data is structured by **unique IDs** to improve scalability and retrieval efficiency

Users


```
users
  0mbXoRmPFKRgCObjBtFJIOJRL8q2
  3OcjhveZ3Bd0E6d6X7A4Gqf6sjn1
  7ujC2FwEsjdwFN4KbUYLrgW2YqQ2
    address: "mm"
    age: "12"
    badgeLevel: 2
    bio: ""
    booksDonated: 1
    booksExchanged: 7
    credits: 1
    email: "yeonghuini@gmail.com"
    genrePreference: "Fantasy"
    phone: "011"
    totalPoints: 95
    userId: "7ujC2FwEsjdwFN4KbUYLrgW2YqQ2"
    username: "yuni"
```


Books


```
-OhxC81dEy1T6J8AY5Sh
  author: "byebye"
  condition: "New"
  description: "bellooooooo"
  genre: "Mystery"
  id: "-OhxC81dEy1T6J8AY5Sh"
  listingType: "Donation"
  ownerId: "ALTS7pyjmJh9EGPNdRoWAUXqTJM2"
  stability: 0
  title: "byebye"
```


Posts


```
-Oi3fCGLZu0WWwkdO_ev
  commentCount: 0
  message: "hihihi"
  postType: "Review"
  posterId: "dP3Ve2hxHvglyJGMdd7ulnzs78u2"
  posterName: "Demo3"
  timestamp
  title: "buaibuai"
```


PointsTransactions


```
-Ohx6Y5IAd9asICb7omz
  points: 10
  timestamp
  type: "Donation"
  userId: "ALTS7pyjmJh9EGPNdRoWAUXqTJM2"
```

Verification of Non-Functional Requirements

Usability

- Real-world user testing on core features
- Tasks completed with **minimal guidance**
- Consistent UI and theme options improve accessibility

Performance

- Main screens load within **~2 seconds**
- Real-time updates for listings and transactions

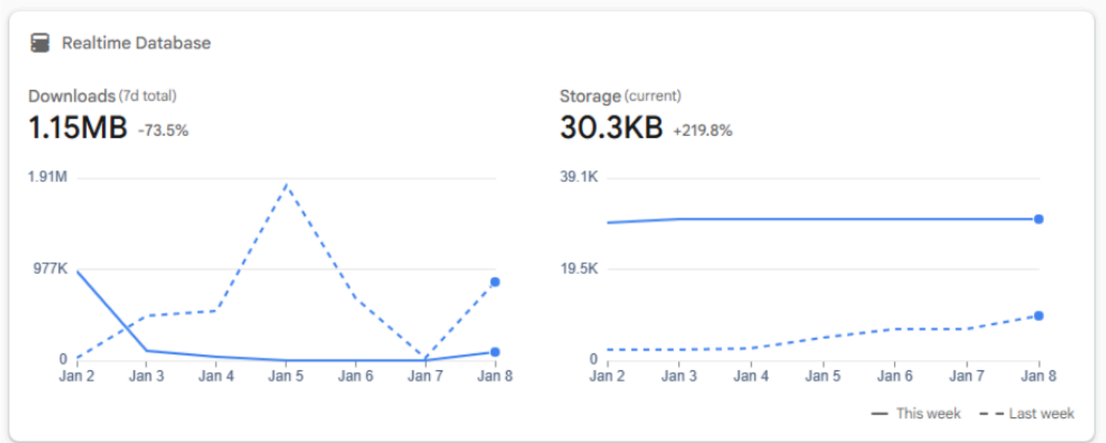
Page Load Time

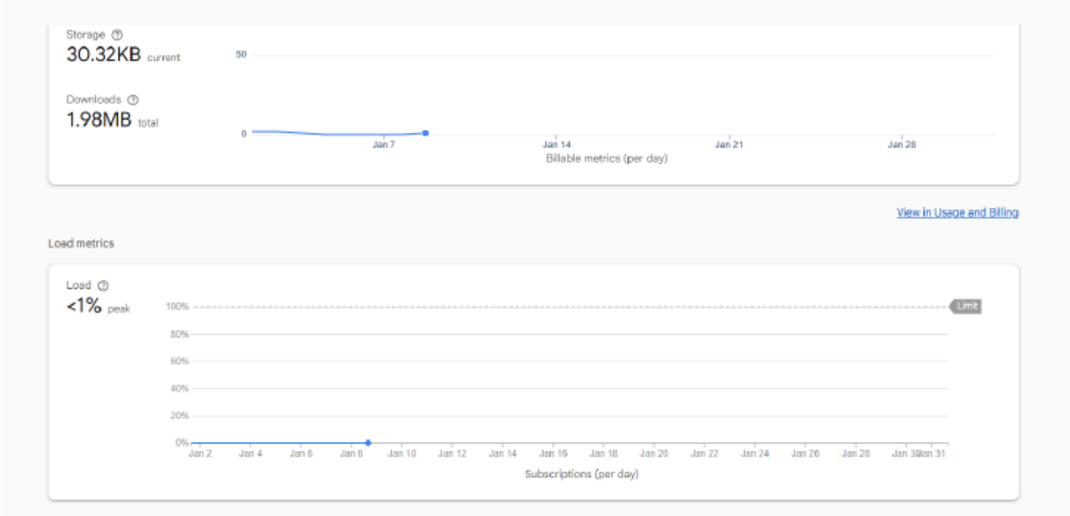
Scalability

- Stable performance with up to **500 concurrent users**
- No feature degradation during concurrent access

Concurrent Users

Verification of Non-Functional Requirements





Evidence

- **User Testing Sessions**
  - Documented feedback and task completion rates.
  -
- **Firebase Monitoring**
  - Real-time data on app performance and stability.
  -
- **Real Device Testing**
  - Validation across various Android devices for optimal compatibility.

Team Contributions

Members	Contributions
Zi Xuan	Lead project, GitHub merge and Module 3 (Recommendation).
Chee Yan	User study, UI design, and Module 1 (Auth).
Hui Ni	UI design, Module 6 (Eco-Impact), and Lead Tester.
Kang Zheng	Firebase database, UI design and Module 4 (Donation/Exchange).
Qian Yi	Firebase database, UI design, and Module 2 (Marketplace).
Yu Xuan	Workflow design, UI design, and Module 5 (Community Forum).

Summary

Key Achievements

- Successfully developed **BookCare**, a mobile app promoting sustainable book reuse
- Integrated **eco-points, recommendations, and community features** aligned with SDG 12.5
- Implemented **secure, scalable, and user-friendly system** using Firebase

Lesson Learned

- Importance of **usability and accessibility** in sustainable app design
- Effective **teamwork, version control**, and task coordination
- **Practical experience** in mobile development and real-world testing

Future Improvements

- In-app messaging between users
- Location-based book matching
- Enhanced recommendation algorithms